TIRANA NOOR FATYANOSA



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GitHub
Medium
LinkedIn
Google Scholar
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SINTA DIKTI
Website

RESEARCH INTERESTS

Language Modeling, Natural Language Processing, Data Science, Meta-Learning, and Evolutionary Algorithm.

EDUCATION

DOCTOR OF PHILOSOPHY, COMPUTER SCIENCE (2021)
Kumamoto University, Japan

MASTER'S DEGREE, COMPUTER SCIENCE (2017)
Brawijaya University, Indonesia

BACHELOR'S DEGREE, COMPUTER SCIENCE (2013)
Brawijaya University, Indonesia

EXPERIENCE

AI ENGINEER / PEDIAEXPERT
Jul 2025 – Present

ASSISTANT PROFESSOR / BRAWIJAYA UNIVERSITY
Mar 2023 – Present

DATA SCIENCE CONSULTANT / KUMAMOTO UNIVERSITY
Sep 2022 – Mar 2024

RESEARCHER / KUMAMOTO UNIVERSITY
Sep 2021 – Aug 2022

AI RESEARCH SCIENTIST INTERN / KATA.AI
Nov 2020 – Apr 2021

QUALITY ASSURANCE SPECIALIST / ANOMADIC
Jul 2017 – Jul 2018

RESEARCH ASSISTANT / BRAWIJAYA UNIVERSITY
Feb 2016 – Aug 2017

SOFTWARE ENGINEER / MITRAIS
Jun 2013 – Feb 2016

AWARDS

TOP 10 FINALIST AT THE BUDAYAGO! GRAND FINAL (PROFESSIONAL CATEGORY)
Dec 2025
Top 10 Finalist position at the BudayaGO! Grand Final in the Professional Category as a member of Team Signesia.

BEST PAPER
Oct 2025
Best Paper at Information Technology International Seminar (ITIS).

HIGHEST SCORER OF AI-900 CERTIFICATION
June 2025
Highest Scorer of AI-900 Certification for Faculty Master Trainer at Universitas Brawijaya.

BEST PAPER
Oct 2024
Best Paper at International Conference on Sustainable Information Engineering and Technology (SIET).

BEST PAPER
Oct 2024
Best Paper at Seminar Nasional Teknologi dan Rekayasa Informasi (SENTRIN).

ACADEMIC EXCELLENCE AWARD
Mar 2022
Recognition for excellent academic performance issued by Kumamoto University.

MONBUKAGAKUSHO (MEXT) SCHOLARSHIP AWARDEE
Sep 2018
Issued by the Japanese Ministry of Education, Culture, Sports, Science and Technology.

BEST SESSION PRESENTER
Oct 2017
Best Session Presenter at the International Conference on Advanced Computer Science and Information Systems (ICACSIS).

BEST PRESENTER
Oct 2016
Best Presenter at the International Conference on Sustainable Information Engineering and Technology (SIET).

PUBLICATIONS

[1] Lailil Muflikhah, Wahyu Kusuma Wardhana, Tirana Noor Fatyanosa, Rizal Setya Perdana, Hana Ratnawati, Nashi Widodo, Solimun Solimun. 2026. Optimizing deep neural network using genetic algorithm for hypertension risk prediction on mobile apps. Array.

[2] Nisma Fadillah Abdul Rahman Onge; Tirana Noor Fatyanosa; Irawati Nurmala Sari. 2026. Data Augmentation and Hyperparameter Optimization for Indonesian Sign Language Gloss Translation Using IndoBART. 2026 22nd IEEE International Colloquium on Signal Processing & Its Applications (CSPA).
[3] Salsa Zufar Radinka Akmal; Naufal Afkaar; Ananda Ravi Kuntadi; Irawati Nurmala Sari; Tirana Noor Fatyanosa. 2026. A Systematic Comparison of MediaPipe Holistic and RTMPose Frameworks for Indonesia Sign Language Skeleton Extraction. 2026 22nd IEEE International Colloquium on Signal Processing & Its Applications (CSPA).
[4] Dayang Alyssa Raisaputri; Tirana Noor Fatyanosa; Irawati Nurmala Sari. 2026. Data-Augmented Text to Gloss Translation for Indonesian Sign Language using Parameter-Efficient FLAN-T5. 2026 22nd IEEE International Colloquium on Signal Processing & Its Applications (CSPA).
[5] Alfih Nadiansyah, Shidiq Al Hakim, Al Hafiz Akbar Maulana Siagian & Tirana Noor Fatyanosa. 2025. Design and implementation of a 360-degree panoramic wedding package booking system using user-centered-design. International Journal of Information Technology
[6] Muhammad Rizqi Azhari, Tirana Noor Fatyanosa, Candra Dewi. 2025. Fine-Tuning Deep Convolutional Generative Adversarial Networks (DCGAN) for Synthetic Retinal Fundus Image Generation. 2025 IEEE 11th Information Technology International Seminar (ITIS).
[7] Danung Nur Adli, Tirana Noor Fatyanosa, Fais AlHuda, Mohammad Miftakhus Sholikin, Sugiharto Sugiharto. 2025. Modelling the growth performance and thermal environment of broiler chicken houses via different machine learning algorithms assisted by a customized Internet of Things. Smart Agricultural Technology.
[8] Tirana Noor Fatyanosa, Putra Pandu Adikara, Rochmanu Erfitra, Muhammad Dikna, Sari Dewi Budiwati, Cahyana Cahyana. 2025. UB_Tel-U at SemEval-2025 Task 11: Emotions Without Borders-A Unified Framework for Multilingual Classification Using Augmentation and Ensemble. International Workshop on Semantic Evaluation (SemEval-2025).
[9] Samuel Cahyawijaya, Holy Lovenia, Joel Ruben Antony Moniz, Tack Hwa Wong, Mohammad Rifqi Farhansyah, Thant Thiri Maung, Frederikus Hudi, David Anugraha, Muhammad Ravi Shulthan Habibi, Muhammad Reza Qorib, Amit Agarwal, Joseph Marvin Imperial, Hitesh Laxmichand Patel, Vicky Feliren, Bahrul Ilmi Nasution, Manuel Antonio Rufino, Genta Indra Winata, Rian Adam Rajagede, Carlos Rafael Catalan, Mohamed Fazli Imam, Priyaranjan Pattnayak, Salsabila Zahirah Pranida, Kevin Pratama, Yeshil Bangera, Adisai Na-Thalang, Patricia Nicole Monderin, Yueqi Song, Christian Simon, Lynnette Hui Xian Ng, Richardy Lobo' Sapan, Taki Hasan Rafi, Bin Wang, Supryadi, Kanyakorn Veerakanjana, Piyalitt Ittichaiwong, Matthew Theodore Roque, Karissa Vincentio, Takdanai Kreangphet, Phakphum Artkaew, Kadek Hendrawan Palgunadi, Yanzhi Yu, Rochana Prih Hastuti, William Nixon, Mithil Bangera, Adrian Xuan Wei Lim, Aye Hninn Khine, Hanif Muhammad Zhafran, Teddy Ferdinan, Audra Aurora Izzani, Ayushman Singh, Evan, Jauza Akbar Krito, Michael Anugraha, Fenal Ashokbhai Ilasariya, Haochen Li, John Amadeo Daniswara, Filbert Aurelian Tjiaranata, Eryawan Presma Yulianrifat, Can Udomcharoenchaikit, Fadil Risdian Ansori, Mahardika Krisna Ihsani, Giang Nguyen, Anab Maulana Barik, Dan John Velasco, Rifo Ahmad Genadi, Saptarshi Saha, Chengwei Wei, Isaiah Flores, Kenneth Ko Han Chen, Anjela Gail Santos, Wan Shen Lim, Kaung Si Phyo, Tim Santos, Meisyarah Dwiastuti, Jiayun Luo, Jan Christian Blaise Cruz, Ming Shan Hee, Ikhlasul Akmal Hanif, M.Alif Al Hakim, Muhammad Rizky Sya'ban, Kun Kerdthaisong, Lester James V. Miranda, Fajri Koto, Tirana Noor Fatyanosa, Alham Fikri Aji, Jostin Jerico Rosal, Jun Kevin, Robert Wijaya, Onno P. Kampman, Ruochen Zhang, Börje F. Karlsson, Peerat Limkonchotiwat. 2025. Crowdsource, Crawl, or Generate? Creating SEA-VL, a Multicultural Vision-Language Dataset for Southeast Asia. Association for Computational Linguistics.
[10] Ariel Zakly Pratama, Arwin Marinta, Bagus Triyudanto, Saman Muhamad, Tirana Noor Fatyanosa. 2025. Retrieval-Augmented Generation for Indonesian Criminal Law Information Using the LLaMA Model. Innovative Informatics and Artificial Intelligence Research.
[11] Rizky Dwi Purnomo; Tirana Noor Fatyanosa; Mahendra Data. 2025. A Comparative Analysis of the Impact of Data Preprocessing and Fine-Tuning Methods on BERT's Performance in Text Classification. International Conference on Natural Language Processing (ICNLP).
[12] Lailil Muflikhah, Tirana Noor Fatyanosa, Nashi Widodo, Rizal Setya Perdana, Solimun, Hana Ratnawati. 2025. Feature Selection for Hypertension Risk Prediction Using XGBoost on Single Nucleotide Polymorphism Data. Healthcare

Informatics Research.
[13] R Ahmad Fattah Saskoro, Novanto Yudistira, Tirana Noor Fatyanosa. 2024. A Detection of AI-Generated Images from Various Generators Using Gated Expert Convolutional Neural Network. IEEE Access.
[14] Israel Mendonça dos Santos, Tirana Noor Fatyanosa, Masayoshi Aritsugi, Pedro Henrique González Silva. 2024. A Hybrid Approach with BRKGA and Data Mining for the Early/Tardy Scheduling Problem. IEEE Congress on Evolutionary Computation (CEC).
[15] Tirana Noor Fatyanosa, Neni Alya Firdausanti, Putu Hangga Nan Prayoga, Minoki Kuriu, Masayoshi Aritsugi, Israel Mendonça dos Santos. 2024. Meta-learning for vessel time series data imputation method recommendation. Expert Systems with Applications.
[16] Tirana Noor Fatyanosa and Mahendra Data. 2023. Hybrid Feature Selection Framework for Building Resource Efficient Intrusion Detection Systems Model in the Internet of Things. *International Conference on Sustainable Information Engineering and Technology (SIET)*.
[17] Tirana Noor Fatyanosa, Mahendra Data, Neni Alya Firdausanti, Putu Hangga Nan Prayoga, Israel Mendonça dos Santos, Masayoshi Aritsugi. 2023. ImputAnom: Anomaly Detection Framework Using Imputation Methods for Univariate Time Series. International Conference on Information Integration and Web Intelligence.
[18] Samuel Cahyawijaya, Holy Lovenia, Alham Fikri Aji, Genta Indra Winata, Bryan Wilie, Rahmad Mahendra, Christian Wibisono, Ade Romadhony, Karissa Vincentio, Fajri Koto, Jennifer Santoso, David Moeljadi, Cahya Wirawan, Frederikus Hudi, Ivan Halim Parmonangan, Ika Alfina, Muhammad Satrio Wicaksono, Ilham Firdausi Putra, Samsul Rahmadani, Yulianti Oenang, Ali Akbar Septiandri, James Jaya, Kaustubh D. Dhole, Arie Ardiyanti Suryani, Rifki Afina Putri, Dan Su, Keith Stevens, Made Nindyatama Nityasya, Muhammad Farid Adilazuarda, Ryan Ignatius, Ryandito Diandaru, Tiezheng Yu, Vito Ghifari, Wenliang Dai, Yan Xu, Dyah Damapuspita, Cuk Tho, Ichwanul Muslim Karo Karo, Tirana Noor Fatyanosa, Ziwei Ji, Pascale Fung, Graham Neubig, Timothy Baldwin, Sebastian Ruder, Herry Sujaini, Sakriani Sakti, Ayu Purwarianti. 2023. NusaCrowd: Open Source Initiative for Indonesian NLP Resources. Association for Computational Linguistics.
[19] Tirana Noor Fatyanosa, Neni Alya Firdausanti, Luis Francisco Japa Soto, Israel Mendonça dos Santos, Putu Hangga Nan Prayoga, Masayoshi Aritsugi. 2023. Conducting Vessel Data Imputation Method Selection Based on Dataset Characteristics. International Seminar on Ocean, Coastal Engineering, Environmental and Natural Disaster Management.
[20] Israel Mendonça dos Santos, Mai Miyaura, Tirana Noor Fatyanosa, Daiki Yamaguchi, Hanami Sakai, Hiroki Obata, Masayoshi Aritsugi. 2023. Classification of unexposed potsherd cavities by using deep learning. Journal of Archaeological Science: Reports.
[21] Neni Alya Firdausanti, Tirana Noor Fatyanosa, Mahendra Data, Israel Mendonça, and Masayoshi Aritsugi. 2022. Two-Stage Sampling: A Framework for Imbalanced Classification with Overlapped Classes. *International Conference on Big Data.*
[22] Alham Fikri Aji, Tirana Noor Fatyanosa, Radityo Eko Prasojo, Philip Arthur, Suci Fitriany, Salma Qonitah, Nadhifa Zulfa, Tomi Santoso, and Mahendra Data. 2022. ParaCotta: Synthetic Multilingual Paraphrase Corpora from the Most Diverse Translation Sample Pair. *Pacific Asia Conference on Language, Information and Computation.*
[23] Sari Dewi Budiwati, Tirana Noor Fatyanosa, Mahendra Data, Dedy Rahman Wijaya, Patrick Adolf Telnoni, Arie Ardiyanti Suryani, Agus Pratondo, and Masayoshi Aritsugi. 2021. To Optimize, or Not to Optimize, That Is the Question: TelU-KU Models for WMT21 Large-Scale Multilingual Machine Translation. *Conference on Machine Translation.*
[24] Tirana Noor Fatyanosa and Masayoshi Aritsugi. 2021. An Automatic Convolutional Neural Network Optimization Using a Diversity-Guided Genetic Algorithm. *IEEE Access.*
[25] Alham Fikri Aji, Haryo Akbarianto Wibowo, Made Nindyatama Nityasya, Radityo Eko Prasojo, and Tirana Noor Fatyanosa 2021. BERT Goes Brrr: A Venture Towards the Lesser Error in Classifying Medical Self-Reporters on Twitter. *Social Media Mining for Health (#SMM4H) Workshop and Shared Task.*
[26] Sari Dewi Budiwati, Al Hafiz Akbar Maulana Siagian, Tirana Noor Fatyanosa, and Masayoshi Aritsugi. 2019. Phrase Table Combination Based on Symmetrization of Word Alignment for Low-Resource Languages. *Applied Sciences.*
[27] Tirana Noor Fatyanosa and Masayoshi Aritsugi. 2020. Effects of the Number of Hyperparameters on the Performance

of GA-CNN. *International Conference on Big Data Computing, Applications and Technologies (BDCAT)*.
[28] Sari Dewi Budiwati, Al Hafiz Akbar Maulana Siagian, Tirana Noor Fatyanosa, and Masayoshi Aritsugi. 2019. DBMS-KU Interpolation for WMT 2019 News Translation Task. *Workshop on Statistical Machine Translation (WMT)*.
[29] Tirana Noor Fatyanosa, Al Hafiz Akbar Maulana Siagian, and Masayoshi Aritsugi. 2019. DBMS-KU at SemEval-2019 Task 9: Exploring Machine Learning Approaches in Classifying Text as Suggestion or Non-Suggestion. *International Workshop on Semantic Evaluation (SemEval)*.
[30] Tirana Noor Fatyanosa, Wayan Firdaus Mahmudy, and Marjuki, 2019. Hybrid Modified Evolution Strategies and Linear Programming for Beef Cattle Feed Optimization. *International Journal on Electrical Engineering and Informatics (IJEEI)*.
[31] Tirana Noor Fatyanosa, Fitra A. Bachtiar, and Mahendra Data. 2018. Feature Selection using Variable Length Chromosome Genetic Algorithm for Sentiment Analysis. *International Conference on Sustainable Information Engineering and Technology (SIET)*.
[32] Tirana Noor Fatyanosa, Wayan Firdaus Mahmudy, and Marjuki, 2018. A Comparison of Four Types of Evolution Strategies for Beef Cattle Feed Optimization. *Journal of Telecommunication, Electronic and Computer Engineering (JTEC)*.
[33] Tirana Noor Fatyanosa, Fitri Utaminingrum, and Mahendra Data. 2018. Linear Programming Initialization Method of Evolution Strategies for Beef Cattle Feed Optimization. *Journal of Telecommunication, Electronic And Computer Engineering (JTEC)*.
[34] Tirana Noor Fatyanosa and Fitra A. Bachtiar. 2017. Classification Method Comparison on Indonesian Social Media Sentiment Analysis. *International Conference on Sustainable Information Engineering and Technology (SIET)*.
[35] Tirana Noor Fatyanosa, Gusti Ahmad Fanshuri Alfarisy, Arief Andy Soebroto, Fatwa Ramdani, and Wayan Firdaus Mahmudy. 2017. Extracting Fuzzy Rules and Parameters Using Particle Swarm Optimization for Rainfall Forecasting. *International Conference on Advanced Computer Science and Information Systems (ICACSIS)*.
[36] Gusti Ahmad Fanshuri Alfarisy, Andreas Nugroho Sihananto, Tirana Noor Fatyanosa, M. Shochibul Burhan, and Wayan Firdaus Mahmudy. 2016. Hybrid Genetic Algorithm and Simulated Annealing for Function Optimization. *Journal of Information Technology and Computer Science (JITECS)*.
[37] Tirana Noor Fatyanosa and Wayan Firdaus Mahmudy. 2016. Implementation of Real Coded Genetic Fuzzy System for Rainfall Forecasting. *International Conference on Sustainable Information Engineering and Technology (SIET)*.